Practice construction of confidence intervals, using the pacemaker dataset

In these notes, we will practice constructing confidence intervals in R using the pacemaker dataset. We will start by loading the necessary libraries and data.

```r
library(tidyverse)
library(krulRutils)
library(here)
library(GGally)
library(janitor)
library(patchwork)

pacemaker_tbl <- read_csv(
  here("data", "pacemaker.csv")
) %>%
  clean_names() %>%
  drop_na() %>%
  rename(
    period = periodo_entre_pulsos,
    intensity = intensidad_de_pulso,
  )
```

This dataset is supposed to contain information about a new pacemaker that is yet to hit the market. The purpose of this report is to analyse the information given and conclude whether the pacemaker provides any significant benefits to the patients. We start by constructing a lookup table for the "marcapasos" variable, and converting it into a factor variable.

```r
pacemaker_lookup_tbl <- tibble(
  code = c("Sin MP", "Con MP"),
  label = c("Without Pacemaker", "With Pacemaker")
)
```

```r
pacemaker_factor <- pacemaker_tbl %>%
  convert_codes_to_factor(
    code_col = marcapasos,
    lookup_tbl = pacemaker_lookup_tbl,
    lookup_code_col = code,
    lookup_label_col = label,
    new_factor_col_name = pacemaker
  ) %>%
  select(-marcapasos)
```

1 Analysis of the "period" variable

We will start by focusing on the "period" variable, which represents the time between pulses in milliseconds.

1.1 Histogram of the "period" variable

To start our analysis, we will create a histogram of the "period" variable for both groups of patients: those with a pacemaker and those without. This will help us visualize the distribution of the "period" variable in each group.

```r
pacemaker_period_histogram <- pacemaker_factor %>%
  ggplot(aes(x = period, fill = pacemaker)) +
  facet_wrap(~ pacemaker, ncol = 1) +
  geom_histogram(
    bins = 30,
    color = "black",
    alpha = 1
  ) +
  c_scale_fill("C blue", "C violet") +
  labs(
    title = "Histograms of Period Variable",
    x = "Period (ms)",
    y = "Count",
    fill = "Pacemaker"
  ) +
  theme_krul() +
  theme(
    legend.position = "none"
  )
```

```
pacemaker_period_histogram
```

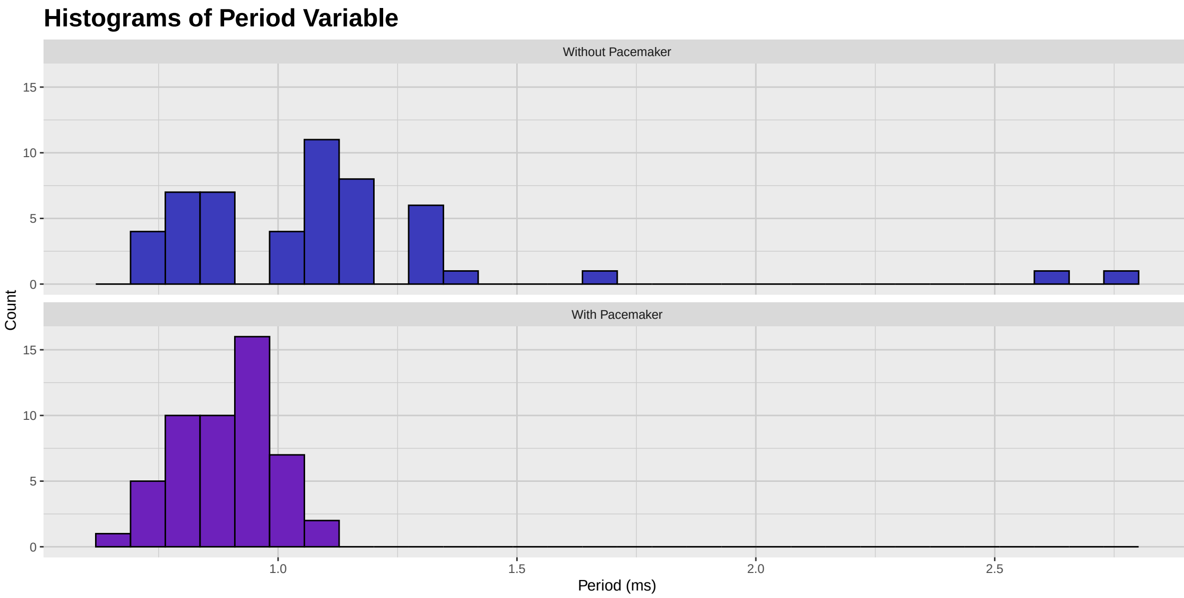

Figure 1: Histogram of the period variable for both groups of patients

As we can see, the patients with the pacemaker have a more uneven distribution of the "period" variable, while the patients without the pacemaker have a more normal distribution. In particular, there are quite a number of outliers in the "With Pacemaker" group, which might not be beneficial for the patients. We will check this by looking at the boxplots of the "period" variable for both groups of patients.

1.2 Boxplot of the "period" variable

As we mentioned, we will now create boxplots of the "period" variable for both groups of patients, this will help us visualize the distribution of this variable and identify any potential outliers.

```
pacemaker_period_boxplot <- pacemaker_factor %>%
  ggplot(aes(x = period, y = 0, fill = pacemaker, color = pacemaker)) +
  facet_wrap(~pacemaker, ncol = 1) +
  geom_boxplot(
    color = "black",
    outlier.shape = NA,
    alpha = 1
```

```
  ) +
  stat_boxplot(
    geom = "errorbar",
    color = "black",
    width = 0.3,
    linewidth = 0.5
  ) +
  geom_jitter(
    alpha = 0.8,
    size = 0.8,
    height = 0.3
  ) +
  c_scale_fill("C blue", "C violet") +
  c_scale_color("C rose", "C red") +
  labs(
    title = "Boxplots of Period Variable",
    x = "Period (ms)",
    y = ""
  ) +
  theme_krul() +
  theme(
    legend.position = "none",
    axis.text.y = element_blank(),
    axis.ticks.y = element_blank()
  )

pacemaker_period_boxplot
```

Boxplots of Period Variable

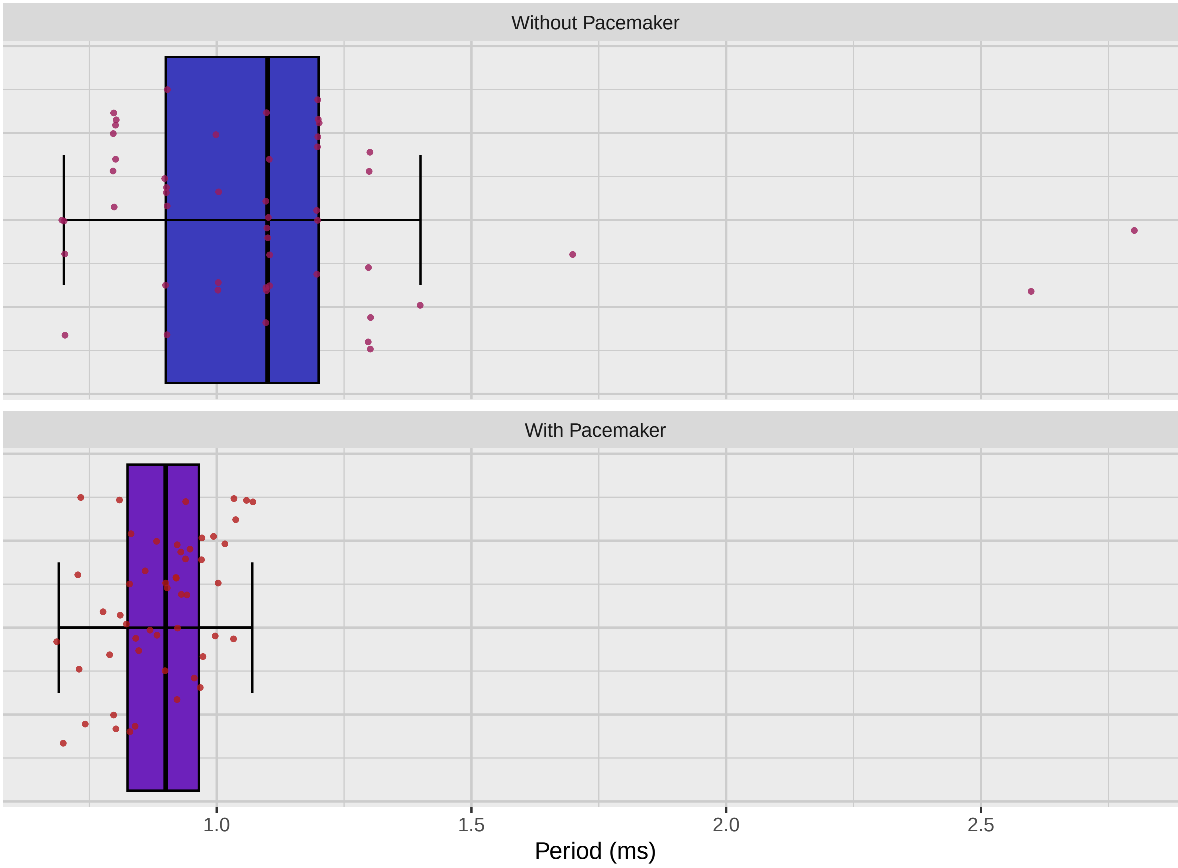

Figure 2: Boxplot of the period variable for both groups of patients

Notice that this plot confirms our previous observation that the "With Pacemaker" group has a more uneven distribution of the "period" variable, with some outliers that might not be beneficial for the patients.

1.3 Confidence intervals for the mean of the "period" variable

We will now construct the confidence intervals, with a confidence level of 0.95, for the mean of the "period" variable for both groups of patients. We start by creating a tibble containing all the relevant information, including the mean and the confidence intervals for each group.

```
pacemaker_period_ci <- pacemaker_factor %>%
  group_by(pacemaker) %>%
  summarise(
```

```r
    x_bar = mean(period),
    ci_lower = t.test(period, conf.level = 0.95)$conf.int[1],
    ci_upper = t.test(period, conf.level = 0.95)$conf.int[2],
  ) %>%
  ungroup()
```

With this information, we can now create a plot showing the confidence intervals for the mean of the "period" variable for both groups of patients.

```r
pacemaker_period_ci_plot <- pacemaker_period_ci %>%
  ggplot(aes(x = x_bar, y = pacemaker, xmin = ci_lower, xmax = ci_upper)) +
  geom_errorbar(
    width = 0.1,
    color = c_pal("C blue"),
    linewidth = 1
  ) +
  geom_point(
    size = 3,
    color = c_pal("C red")
  ) +
  labs(
    title = "Confidence Intervals for Mean of Period Variable",
    x = "Mean Period (ms)",
    y = ""
  ) +
  theme_krul()

pacemaker_period_ci_plot
```

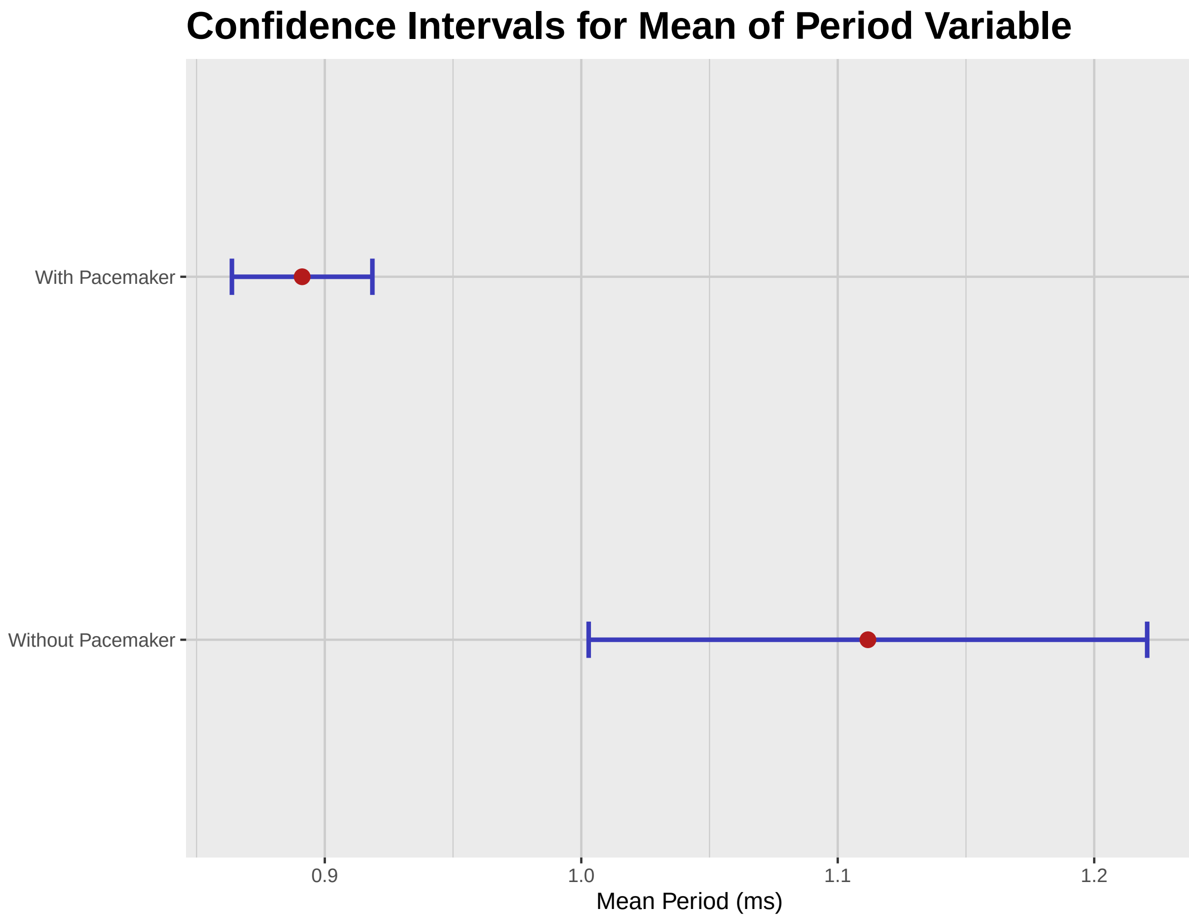

Figure 3: Confidence intervals for the mean of the period variable for both groups of patients

From this plot, we can see that the confidence intervals for the mean of the “period” variable for both groups of patients do not overlap. This suggests that there is a significant difference between the two groups, and that the pacemaker does have an effect on the “period” variable. However, it is not clear if the slower period induced by the pacemaker is beneficial for the patients, specially taking into account the outliers we observed in the previous plots.

2 Analysis of the “intensity” variable

We will now focus on the “intensity” variable, which represents the intensity of the pulse in milliamperes.

2.1 Histogram of the "intensity" variable

To start our analysis, we will create a histogram of the "intensity" variable for both groups of patients: those with a pacemaker and those without. This will help us visualize the distribution of the "intensity" variable in each group.

```r
pacemaker_intensity_histogram <- pacemaker_factor %>%
  ggplot(aes(x = intensity, fill = pacemaker)) +
  facet_wrap(~ pacemaker, ncol = 1) +
  geom_histogram(
    bins = 30,
    color = "black",
    alpha = 1
  ) +
  c_scale_fill("C blue", "C violet") +
  labs(
    title = "Histograms of Intensity Variable",
    x = "Intensity (mA)",
    y = "Count",
    fill = "Pacemaker"
  ) +
  theme_krul() +
  theme(
    legend.position = "none"
  )

pacemaker_intensity_histogram
```

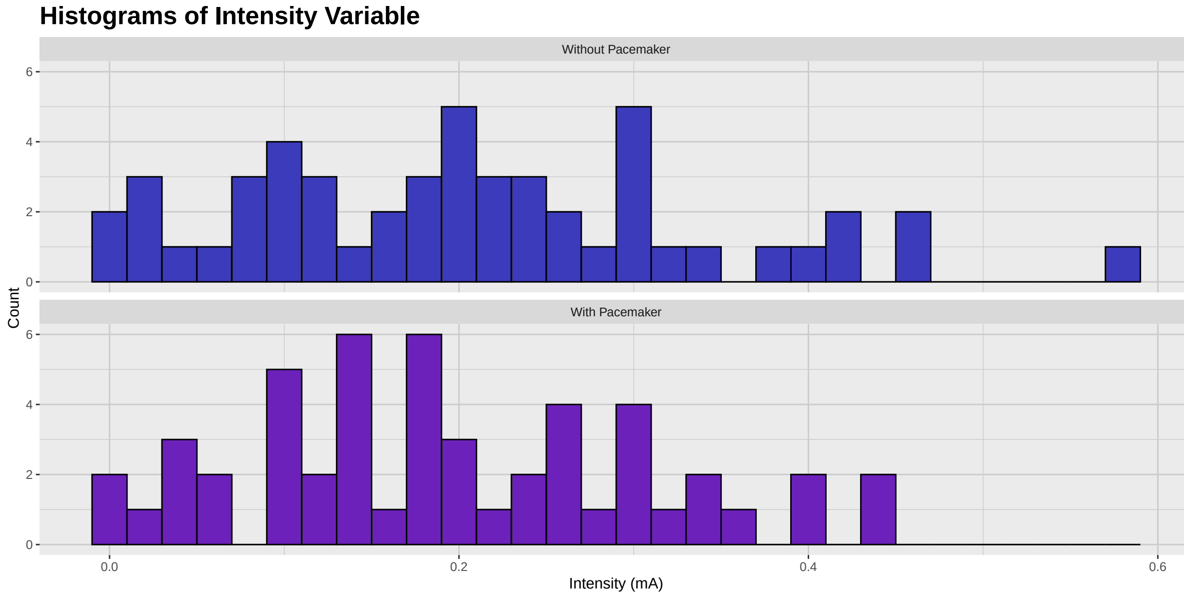

Figure 4: Histogram of the intensity variable for both groups of patients

As we can see, the patients with the pacemaker have a more uneven distribution of the "intensity" variable, while the patients without the pacemaker have a more normal distribution. In particular, there are quite a number of outliers in the "With Pacemaker" group, which might not be beneficial for the patients. We will check this by looking at the boxplots of the "intensity" variable for both groups of patients.

2.2 Boxplot of the "intensity" variable

As we mentioned, we will now create boxplots of the "intensity" variable for both groups of patients, this will help us visualize the distribution of this variable and identify any potential outliers.

```
pacemaker_intensity_boxplot <- pacemaker_factor %>%
  ggplot(aes(x = intensity, y = 0, fill = pacemaker, color = pacemaker)) +
  facet_wrap(~pacemaker, ncol = 1) +
  geom_boxplot(
    color = "black",
    outlier.shape = NA,
    alpha = 1
  ) +
  stat_boxplot(
    geom = "errorbar",
```

```
    color = "black",
    width = 0.3,
    linewidth = 0.5
  ) +
  geom_jitter(
    alpha = 0.8,
    size = 0.8,
    height = 0.3
  ) +
  c_scale_fill("C blue", "C violet") +
  c_scale_color("C rose", "C red") +
  labs(
    title = "Boxplots of Intensity Variable",
    x = "Intensity (mA)",
    y = ""
  ) +
  theme_krul() +
  theme(
    legend.position = "none",
    axis.text.y = element_blank(),
    axis.ticks.y = element_blank()
  )

pacemaker_intensity_boxplot
```

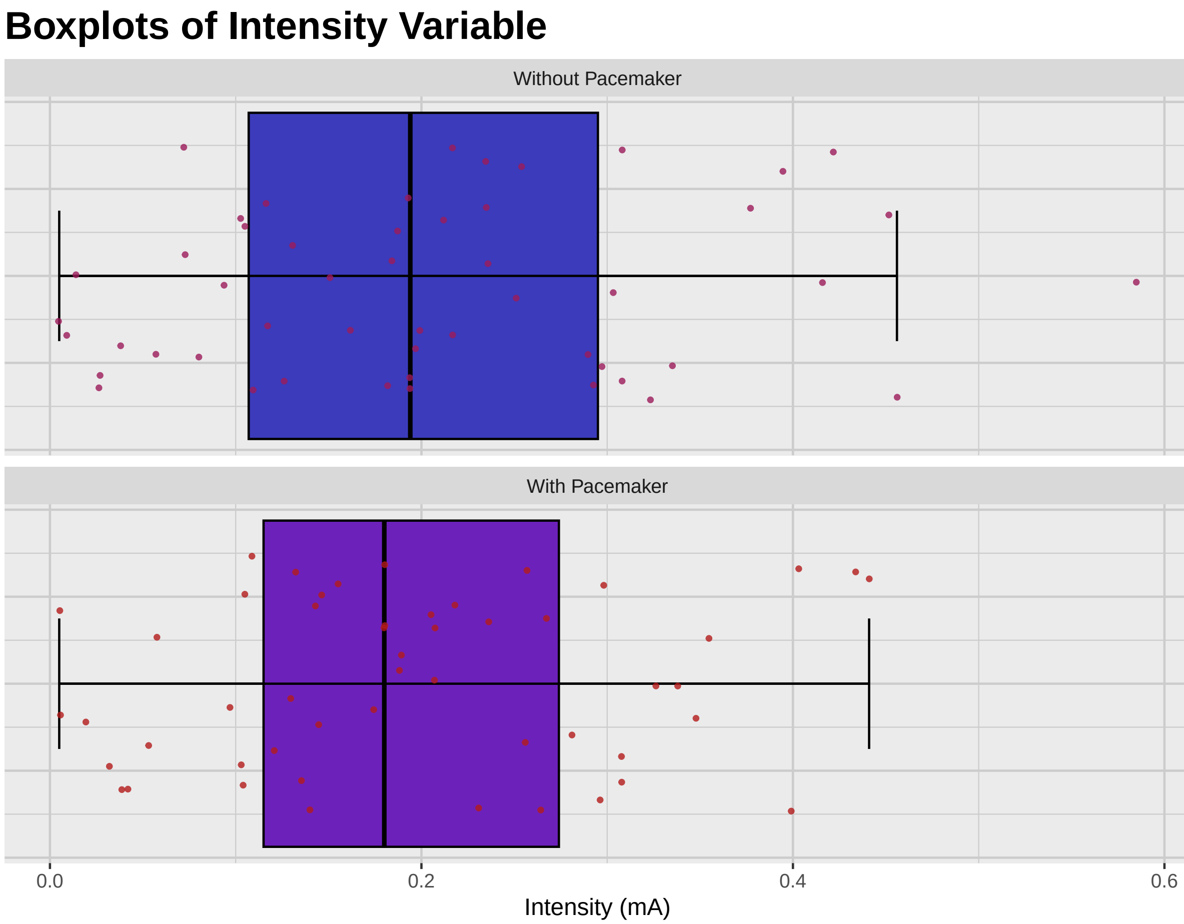

Figure 5: Boxplot of the intensity variable for both groups of patients

Notice that this plot confirms our previous observation that the "With Pacemaker" group has a more uneven distribution of the "intensity" variable, with some outliers that might not be beneficial for the patients.

2.3 Confidence intervals for the mean of the "intensity" variable

We will now construct the confidence intervals, with a confidence level of 0.95, for the mean of the "intensity" variable for both groups of patients. We start by creating a tibble containing all the relevant information, including the mean and the confidence intervals for each group.

```
pacemaker_intensity_ci <- pacemaker_factor %>%
  group_by(pacemaker) %>%
  summarise(
```

```r
    x_bar = mean(intensity),
    ci_lower = t.test(intensity, conf.level = 0.95)$conf.int[1],
    ci_upper = t.test(intensity, conf.level = 0.95)$conf.int[2],
  ) %>%
  ungroup()
```

With this information, we can now create a plot showing the confidence intervals for the mean of the "intensity" variable for both groups of patients.

```r
pacemaker_intensity_ci_plot <- pacemaker_intensity_ci %>%
  ggplot(aes(x = x_bar, y = pacemaker, xmin = ci_lower, xmax = ci_upper)) +
  geom_errorbar(
    width = 0.1,
    color = c_pal("C blue"),
    linewidth = 1
  ) +
  geom_point(
    size = 3,
    color = c_pal("C red")
  ) +
  labs(
    title = "Confidence Intervals for Mean of Intensity Variable",
    x = "Mean Intensity (mA)",
    y = ""
  ) +
  theme_krul()

pacemaker_intensity_ci_plot
```

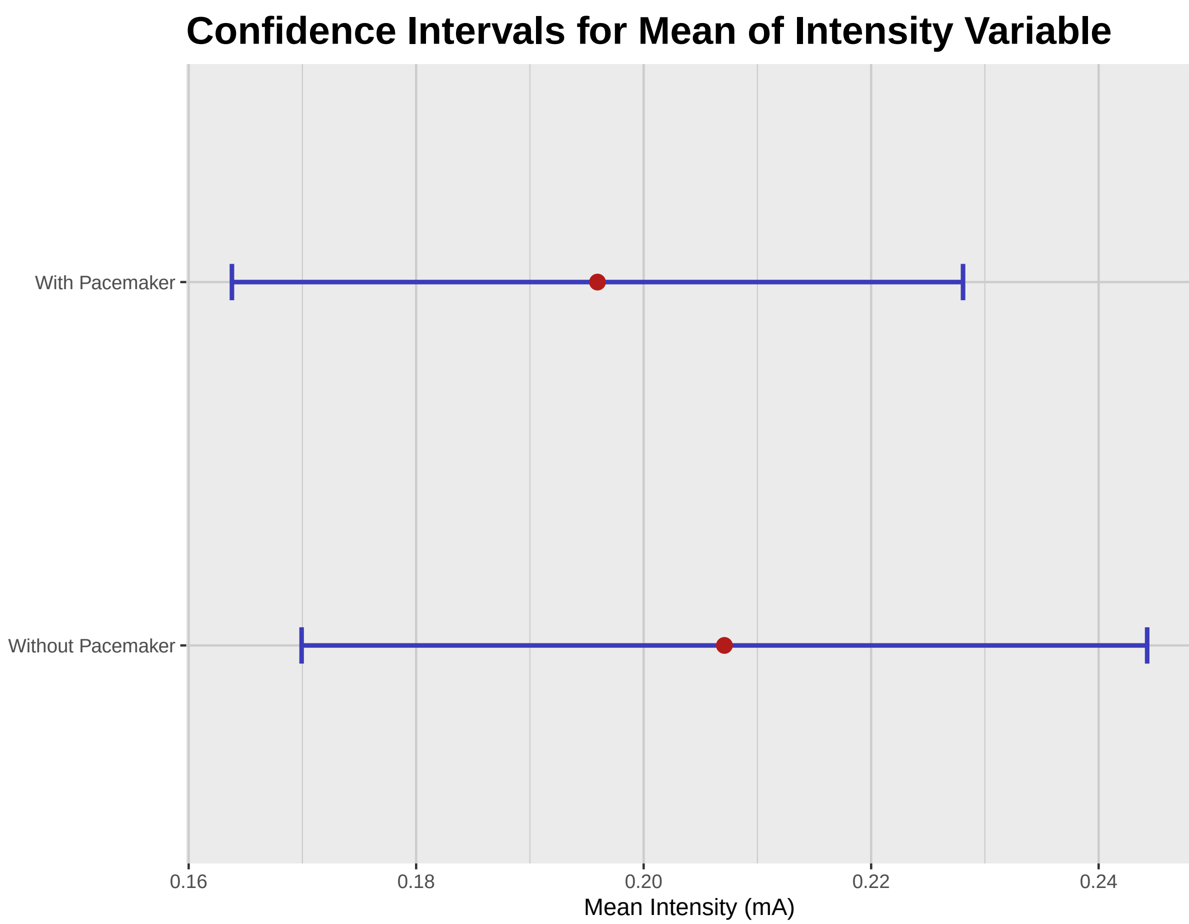

Figure 6: Confidence intervals for the mean of the intensity variable for both groups of patients

From this plot, we can see that the confidence intervals for the mean of the "intensity" variable for both groups of patients overlap quite a bit. This suggests that there is no significant difference between the two groups, and that the pacemaker does not have an effect on the "intensity" variable.

3 Conclusion

Based on the data collected, we can't really recommend the introduction of this new pacemaker into the market. The pacemaker effect on the "intensity" variable is not significant. The effect on the "period" variable is significant, but it is not clear if the slower period induced by the pacemaker is beneficial for the patients, specially taking into account the outliers we observed in the previous plots.